Technical Data Sheet

# ALCOM PA66 910/1.1 CF10

(Last update: 24.08.2022)

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|---|---|
| Base Polymer | Polyamide 66 |
| Filler/Additive System | 10 % carbon fibres |
| Special Features | electrically conductive, reduced surface resistivity, heat stabilised, high stiffness |
| Market Segment | Automotive, Machinery |
| Application Area | various |
| Typical Applications | bearings, functional components |

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| Pre-Drying Conditions | 80 °C in a dry air (dessiccant) dryer<br>for 2-12 h<br>max. moisture content <0,15 % |
| Processing Injection Moulding | melt temperature 280-300 °C<br>mould temperature 80-120 °C |
| Storage | dry, protected from light |

| Properties | dry/cond. | Dimension | Test Norm |
|---|---|---|---|
| **Mechanical Properties** | | | |
| Flexural Modulus | 7400 / - | MPa | ISO 178 |
| Flexural Strength | 220 / - | MPa | ISO 178 |
| Tensile Modulus | 8100 / - | MPa | ISO 527 |
| Tensile Strength at Break | 155 / - | MPa | ISO 527 |
| Tensile Elongation at Break | 2.4 / - | % | ISO 527 |
| Impact Strength (Charpy, 23°C) | 30 / - | kJ/m² | ISO 179/1eU |
| Impact Strength (Charpy, -40°C) | 28 / - | kJ/m² | ISO 179/1eU |
| Notched Impact Strength (Charpy, 23°C) | 4 / - | kJ/m² | ISO 179/1eA |
| Notched Impact Strength (Charpy, -40°C) | 3 / - | kJ/m² | ISO 179/1eA |
| **Thermal Properties** | | | |
| HDT / A (1,8 MPa) | 250 / * | °C | ISO 75-1/-2 |
| DSC (Melt Point) | 263 / * | °C | ISO 11357 |
| **Electrical Properties** | | | |
| Surface Resistance | * / 700 | Ohm | IEC 62631-3-2 |
| **Rheological Properties** | | | |
| Shrinkage (lengthwise, 24h) | 0.1 - 0.3 | % | ISO 294-4 |
| Shrinkage (lateral, 24h) | 0.6 - 0.8 | % | ISO 294-4 |
| **Physical Properties** | | | |
| Density | 1170 / - | kg/m³ | ISO 1183 |

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## Tribologic Properties

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|---|---|---|---|
| Coefficient of Sliding Friction µ (pv = 5*1 MPa*m/s) | 0.26 | - | ASTM G 137 |
| Coefficient of Sliding Friction µH (pv = 5*1 MPa*m/s) | 0.31 | - | ASTM G 137 |
| Specific Wear Rate ws (pv = 5*1 MPa*m/s) | 0.33 | E-6 mm³/Nm | ASTM G 137 |
| Linear Wear Rate w (pv = 5*1 MPa*m/s) | 5.9 | µm/h | ASTM G 137 |

**Liability Exclusion**
These are guide values and not a specification. The test values mentioned are representative values only and not binding minimum or maximum figures. These test values have been determined on standardised test specimens and can be affected by pigmentation, mould design and processing conditions.

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